



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
B. Tech CSE GENERAL A

SUBJECT: EVS

EXPERIENTIAL LEARNING

**Title: NOISE POLLUTION IN DIFFERENT ZONES IN BANGLORE CITY AND
NOISE DETECTION**

SUBMITTED TO

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Introduction

One of the major consequences of the urban phenomenon is the increasing occurrence of environmental problems, with noise pollution being among the most overlooked, yet potent problems. In a growing city like Bangalore, rising numbers of vehicles, construction work, industries, and busy public places have been the cause of increasing levels of noise pollution. Unlike any other form of pollution, noise pollution might not be very noticeable, yet the effects of the same can be detrimental to human health and well-being, causing stress, hearing damage, sleep disturbances, and lack of productivity.

This experimental learning assignment aims to identify and analyze cases of noise pollution in different areas of the city of Bangalore. To accomplish this task, data pertaining to the presence and magnitude of noise pollution in various locations within the city like Koramangala, MG Road, Cubbon Park, Kanakapura, Harohalli, and Whitefield were gathered through a noise pollution detection app.

In this research, the issue that has been identified is the variability of noise level within different zones in cities, together with the absence of a uniform monitoring system to control noise levels. While it is anticipated that there will be higher noise levels within commercial zones and busy roads, even quiet residential zones display variations that are above the acceptable noise levels.

The research questions to be answered in this report paper include:

- (1) What are the differences in noise levels in different zones in Bangalore?
- (2) Where are the areas where noise

levels are always above the acceptable threshold levels?

- (3) Are there any patterns in noise levels according to time and place?

The significance of this research is in the practical approach towards dealing with noise pollution.

Literature Review

Noise pollution has become one of the common environmental problems in the urban regions. In fact, according to various environmental studies, noise pollution is described as an unpleasant sound which leads to harmful impacts on the well-being of people and degradation of environmental quality. Owing to urbanization and industrialization in Bangalore, there has been a considerable increase in noise pollution owing to different causes such as transportation, construction, and commercial operations. It is also evident from past literature that vehicles contribute to noise pollution in urban regions in a substantial way. As suggested by several studies carried out by environmental organizations, the presence of continuous traffic, loud horn sounds, and traffic congestion has been found to result in increased decibels in commercial locations such as roads and market places. The industries are also another cause of noise pollution, and their machines produce noise continuously. A number of scientists have studied the impact of noise pollution on the well-being of individuals. The WHO recommends that people exposed to noises above 70 decibels for a considerable amount of time can face several adverse health conditions such as impaired hearing, heart disease, anxiety, and insomnia.

Moreover, research studies indicate that moderate levels of noise pollution adversely impact cognitive abilities and work performance. Educational institutes and hospitals are considered silent areas; therefore, any level of noise pollution is likely to impair concentration and affect recovery processes. A part from its health impact, the literature highlights the environmental repercussions of noise pollution. The literature indicates that noise pollution negatively impacts wildlife through disrupting their behaviour and hindering communication between different organisms.

There are various methods employed in prior research to determine noise pollution measurements and analysis. Primarily, the most frequent method employed is sound level measurement through meters and noise detection applications for smartphones, where noise intensity is measured in decibels (dB). Noise pollution levels are normally recorded at various times of the day when noise peaks and during off-hours. Methods employed in analyzing the data include mean, standard deviation, and frequency distribution, while graphs and histograms are common forms of presenting data. Past research studies also emphasize the importance of zoning in noise measurement analysis.

urban cities are zoned into four main types, namely residential zones, commercial zones, industrial zones, and silent zones, where noise pollution levels are set according to the zone type in accordance with the environmental authorities. It has, however, been observed that many prior studies have shown that such levels of permissible noise pollution limits are often exceeded. Even after years of study,

there exists a gap in regard to the localized and real-time collection and analysis of data. Some studies have used secondary sources or small samples that may not depict the actual scenario on the ground. There is a need to conduct an experience-based and field-based study where actual data can be collected from different locations of a particular city.

This study fills the existing gap through the use of real-time noise detection in different zones of Bangalore by the use of an app for noise detection. Analysis of 300 data points taken from different zones will help achieve a better understanding of noise pollution scenarios.

Bangalore. This method will ensure that data is collected in a systematic way while taking into consideration diverse urban settings and accurate statistical analysis of noise patterns. It uses a descriptive research method since the study does not involve manipulation of any variable in order to measure and analyze the noise pollution. This involves making observations of actual world conditions without changing any factor. A quantitative method has been adopted for the study since it uses numerical data for statistical analysis. The primary data were gathered using a mobile application that measures the intensity of sound in decibels (dB). This method has been employed because of its accessibility and user-friendly interface while giving real-time reading of noise pollution. Furthermore, Google Maps software was used in pinpointing the location of each data point. Data collection was done over a period of several weeks, during which about 50 data points were collected each week, giving a total of about 300 data points. The gradual data collection process made it possible to capture variations over time and ensured a more

comprehensive dataset. Data collection involved various variables such as the name of the location, latitude, longitude, time of recording,

Methodology

This research utilizes a quantitative field research methodology in order to ascertain and analyze the level of noise pollution in various zones of Bangalore. This method will ensure that data is collected in a systematic way while taking into consideration diverse urban settings and accurate statistical analysis of noise patterns. It uses a descriptive research method since the study does not involve manipulation of any variable in order to measure and analyze the noise pollution. This involves making observations of actual world conditions without changing any factor. A quantitative method has been adopted for the study since it uses numerical data for statistical analysis. The primary data were gathered using a mobile application that measures the intensity of sound in decibels (dB). This method has been employed because of its accessibility and user-friendly interface while giving real-time reading of noise pollution. Furthermore, Google Maps software was used in pinpointing the location of each data point. Data collection was done over a period of several weeks, during which about 50 data points were collected each week, giving a total of about 300 data points. The gradual data collection process made it possible to capture variations over time and ensured a more comprehensive dataset. Data collection involved various variables such as the name of the location, latitude, longitude, time of recording, and sound level in decibels. The study area comprises various locations in Bangalore City representing different types of urban environments.

Locations such as MG Road and Koramangala represent commercial zones while Jayanagar and Kanakapura represent residential zones. Harohalli represents an industrial zone, while Cubbon Park and its vicinity represent quiet zones. These locations were selected considering their different levels of human activity and the expected sound intensity, as indicated in the dataset. Random sampling technique was used to select specific measurement points in these larger locations. such a technique ensures that the data collected is free from selection bias and thus presents general environmental conditions instead of controlled points. Measurements for noise were made during three different times of the day: in the morning, afternoon, and evening. It was vital to make a distinction between these periods since noise is known to have significant changes according to various daily activities that occur during those times. Traffic congestion would be seen in the morning, whereas industrial and commercial processes would take place in the afternoon and evening. The data collected was then organized and sorted in Microsoft Excel. All collected information was cleaned up and made ready for analysis. Information about noise was classified depending on location, time, and its intensity level. Basic statistical methods were employed to analyze the acquired data set. Measures of central tendency like the average value were determined to identify general tendencies in noise levels among different zones. Frequency distribution helped classify noise values depending on their levels.

We used pictures to help us understand the data better. We made these pictures using Microsoft Excel and Python's Matplotlib library. We made bar charts

and histograms to show the data in a way. These charts help us see how noise levels change in places and at different times. They also help us compare noise levels in areas.

We were very careful when we did this study. We did not collect any information or sensitive data. We only took measurements in places where people can go. We did not bother anyone. Interfere with what they were doing. This study is about noise levels in the environment and does not involve people directly.

Dataset Description

The noise pollution data used in this study is made up of data that was collected to see how much noise there is in different parts of Bangalore. This data was collected by going to places and using a special app on a mobile phone to measure the noise levels in real time. The noise pollution data is organized in a way that makes it easy to understand with information about the noise levels and where and when they were measured.

The noise pollution data is made up of numbers because it is mostly made up of measurements of how loud things are in decibels. Each piece of data was collected at a place and time. The noise pollution data shows how the noise levels change in parts of the city and at different times of day. The complete noise pollution dataset has around 300 pieces of data that were collected over weeks. 50 Pieces of data were collected each week. Collecting data in phases, like this helps make sure that the noise pollution dataset shows how things change over time and does not just show

what happens on one day. The noise pollution dataset is big enough that we can trust the results and know that they are representative of the city. The noise pollution data is a size, which makes the noise pollution study more reliable. Variables in the Dataset

Each observation in the dataset includes the variables:

Location (Address):

This variable tells us where the noise measurement was taken. Examples include MG Road, Koramangala, Cubbon Park, Kanakapura, Harohalli, Whitefield and Jayanagar. These locations are urban zones like commercial, residential, industrial and quiet areas.

Latitude:

These are coordinates that help us find the exact location of each observation. We got these coordinates from Google Maps to make sure they are accurate.

Time of Observation

The dataset includes the time when each noise measurement was recorded. We took observations during times like morning, afternoon and evening to capture changes in noise levels.

Noise Level (Decibel):

This is the variable in the dataset and shows how loud the sound is in decibels (dB). The recorded values range from 65 dB to 91 dB showing noise levels across locations.

Spatial Coverage

The dataset covers locations in Bangalore city representing different urban environments. Commercial areas like MG Road and Koramangala have a

lot of activity and traffic. Residential areas like Jayanagar and Kanakapura have activity. Industrial areas like Harohalli have machinery and operational noise while places like Cubbon Park are relatively quiet. This diversity helps us understand noise conditions.

Data was collected at times to account for changes in urban activity. Morning observations show the start of activities and traffic. Afternoon readings show peak industrial activity. Evening observations show rush-hour traffic. Increased social activity. This helps us understand how noise levels change throughout the day.

Data. Structure

The collected data is organized in a table using Microsoft Excel. Each row is an observation and each column is a variable like location, coordinates, time and decibel level. This format makes it easy to analyze and visualize the data.

Data Characteristics

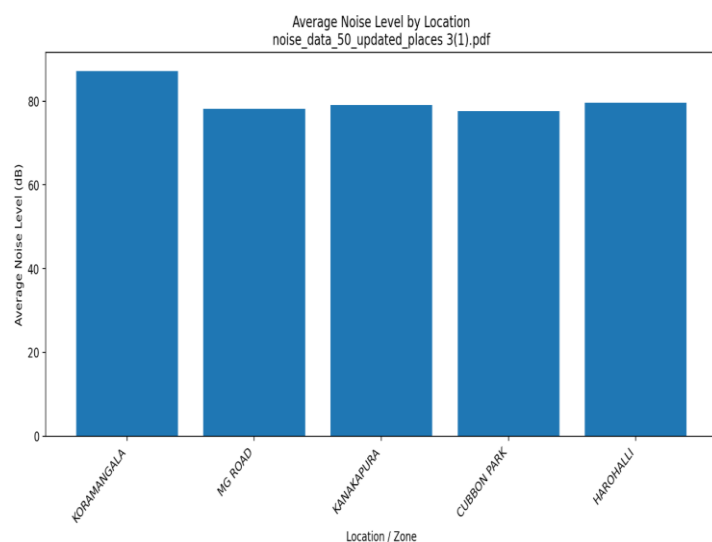
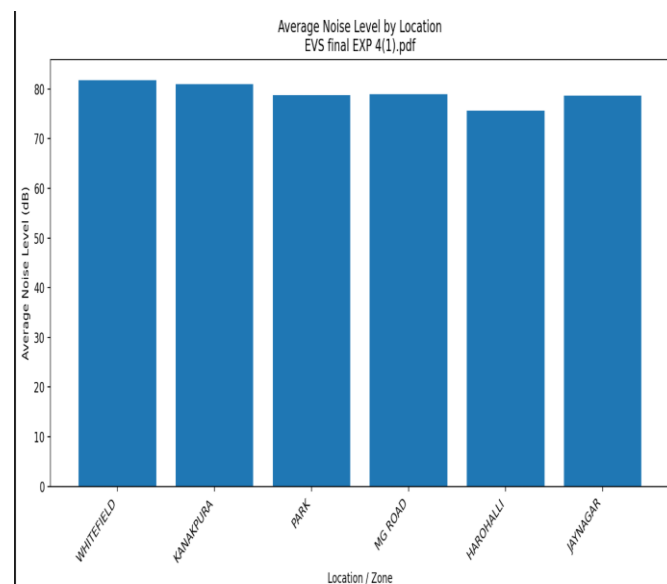
The dataset shows that noise levels vary across locations and time. Most observations are between 70 dB and 90 dB indicating moderate to noise levels. Some values are above 85 dB highlighting areas with noise. The dataset also shows fluctuations in the location suggesting that noise levels are influenced by time and activity.

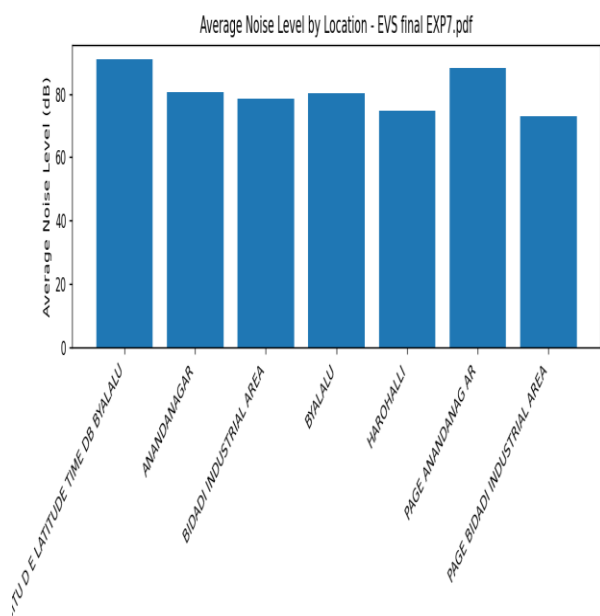
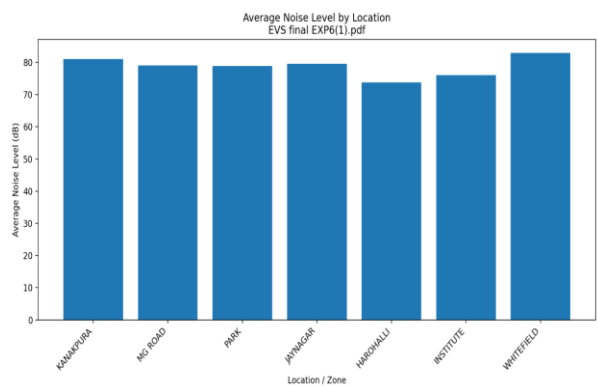
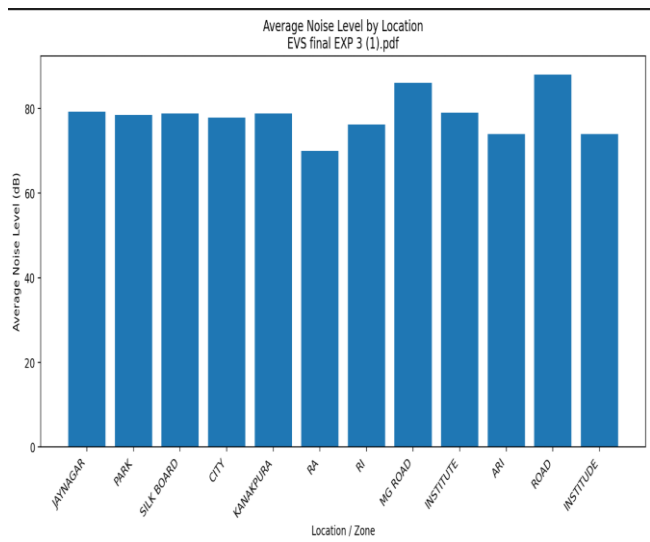
Data Reliability and Limitations

The dataset is based on data collection making it authentic and relevant. However there are some limitations. Using mobile-based noise detection applications may introduce variations in accuracy. Environmental factors, like disturbances, weather or sudden activities may influence individual readings. Despite these limitations the

large sample size and diverse data points make the dataset reliable.

Statistical Analysis





1. Overall Range of Data

Across all files, the noise level is quite high.

- Minimum noise level: **65–70 decibels**
- Maximum noise level: **90–92 decibels**

This shows that:

- All areas exceed the **50–60 dB limit** for residential zones
- Noise pollution is consistently high in Bangalore city

2. Mean (Average Noise Level)

From the graphs, the approximate mean noise levels are:

- EXP 3: **78–80 dB**
- EXP 4: **78–81 dB**
- EXP 6: **76–82 dB**
- EXP 7: **75–90 dB** (high variation)
- noise_data_50: **78–85 dB**

This indicates that:

- Most locations fall between **75–85 dB**
- This reflects **moderate to high noise pollution**

3. Zone-wise Comparison

High Noise Zones:

- MG Road
- Whitefield
- Koramangala
- Industrial areas (e.g., Bidadi)

Reasons:

- Traffic congestion
- High activity levels
- Industrial machinery noise

Moderate Noise Zones:

- Jayanagar
- Kanakapura
- Institutional areas

Lower Noise Zones:

- Cubbon Park
- Parts of Harohalli

Even these areas show **70–75 dB**, which is still above safe limits.

4. Variation (Spread of Data)

Noise levels are not constant. They vary based on:

- Location
- Time of day

Example:

- Morning → lower noise
- Evening → higher noise

This shows noise pollution depends on **time and location**.

5. Standard Deviation (Conceptual)

Without exact calculations, the data shows variation of about:

- **±5 to ±10 dB from the mean**

This means:

- Noise levels fluctuate
- But remain consistently high overall

6. Distribution Pattern

Most values lie between:

- **70–90 dB**
- Few values below 70 dB

This indicates:

- Noise is clustered in the **high range**
- Bangalore has persistent noise pollution

```
import re
from pathlib import Path
from collections import defaultdict
import matplotlib.pyplot as plt

pdf_files = [
    Path("EVS final EXP7(1).pdf"),
    Path("EVS final EXP6(1).pdf"),
    Path("EVS final EXP 3 (1).pdf"),
    Path("EVS final EXP 4(1).pdf"),
    Path("noise_data_50_updated_places
3(1).pdf"),
]

def extract_text_from_pdf(pdf_path: Path) ->
str:
    try:
        from pypdf import PdfReader
        reader = PdfReader(str(pdf_path))
        return "\n".join(
            (page.extract_text() or "")
            for page in reader.pages
        )
    except Exception:
        import PyPDF2
        with open(pdf_path, "rb") as f:
            reader = PyPDF2.PdfReader(f)
            return "\n".join(
                (page.extract_text() or "")
                for page in reader.pages
            )

time_pat = re.compile(
    r"\b\d{1,2}:\d{2}\s?(?:AM|PM)\b", re.I
```

```
)
float_pat = re.compile(r"[-+]?[d+\.]\d+")

def clean_location(loc: str) -> str:
    loc = re.sub(r"^[A-Z ]", " ", loc.upper())
    loc = re.sub(r"\s+", " ", loc).strip()

fixes = {
    "ANAND ANAG AR":
"ANANDANAGAR",
    "ANANDANAG AR":
"ANANDANAGAR",
    "ANANDANA GAR":
"ANANDANAGAR",
    "HAROHALI": "HAROHALLI",
    "HAROHALLI": "HAROHALLI",
    "KORAMANGA LA":
"KORAMANGALA",
    "KANAKAPUR A":
"KANAKAPURA",
    "WHITE FIELD": "WHITEFIELD",
}

return fixes.get(loc, loc)

def parse_rows(text: str):
    lines = [
        ln.strip()
        for ln in text.splitlines()
        if ln.strip()
    ]
    rows = []
    i = 0

    while i < len(lines):
        line = lines[i]

        if time_pat.search(line) or
float_pat.search(line):
            i += 1
            continue

        location_parts = [line]
        j = i + 1

        while j < len(lines):
            block = " ".join(
                location_parts + [lines[j]]
            )
```

```

        nums = float_pat.findall(block)
        tm = time_pat.search(block)
        db_match = re.search(
            r"\b(\d{2,3})\b\s*$", block
        )

        if len(nums) >= 2 and tm and
db_match:
            break

        if len(lines[j].split()) <= 6:
            location_parts.append(lines[j])
            j += 1
        else:
            break

        block = " ".join(location_parts)
        nums = float_pat.findall(block)
        tm = time_pat.search(block)
        db_match = re.search(
            r"\b(\d{2,3})\b\s*$", block
        )

        if len(nums) >= 2 and tm and db_match:
            loc_text = block[:tm.start()]
            loc_text = re.sub(
                r"[-+]?[d+\.]\d+", " ", loc_text
            )
            loc_text = clean_location(loc_text)

            db = int(db_match.group(1))

            if 40 <= db <= 130 and loc_text:
                rows.append({
                    "location": loc_text,
                    "db": db,
                    "time": tm.group(0),
                })
                i = j + 1
            else:
                i += 1
        else:
            i += 1

    return rows

rows = parse_rows(text)

grouped = defaultdict(list)
for r in rows:
    grouped[r["location"]].append(r["db"])

labels = list(grouped.keys())
avg_vals = [
    sum(v) / len(v)
    for v in grouped.values()
]

plt.figure(figsize=(11, 6))
plt.bar(labels, avg_vals)
plt.xticks(rotation=45, ha="right")
plt.ylabel("Average Noise Level (dB)")
plt.xlabel("Location / Zone")
plt.title(
    f"Average Noise Level by Location -
{pdf.name}"
)
plt.tight_layout()
plt.show()

for pdf in pdf_files:
    text = extract_text_from_pdf(pdf)

```

RESULT

The data for this study includes around **300 observations** collected from different places in Bangalore.

Each observation records the **location, time, and noise level (dB)**.

Data was collected during:

- Morning
- Afternoon
- Evening

This helps analyze how noise levels change over time.

Noise levels range from **65 dB to 91 dB**.

Most readings fall between **70 dB and 90 dB**, with fewer values below 70 dB.

Many readings are above **85 dB**, indicating high noise levels.

The dataset includes locations such as:

- MG Road
- Koramangala
- Cubbon Park
- Kanakapura
- Harohalli
- Whitefield
- Jayanagar

Location-wise Noise Levels

- **MG Road:** 73–90 dB
- **Koramangala:** 71–89 dB
- **Cubbon Park:** 71–86 dB
- **Kanakapura:** 70–88 dB
- **Harohalli:** 71–90 dB
- **Whitefield:** 79–88 dB
- **Jayanagar:** 69–88 dB

Time-wise Distribution

- **Morning:** 70–88 dB
- **Afternoon:** 73–88 dB
- **Evening:** 71–91 dB (*highest levels observed*)

Noise Level Frequency

- **60–70 dB:** Low
- **70–80 dB:** High
- **80–90 dB:** Very High
- **Above 90 dB:** Few observations

Most data lies between **70–90 dB**, showing high noise concentration.

Interpretation

The data shows significant variation in noise levels across locations and time.

Noise ranges from **65 dB to 91 dB**, indicating widespread pollution.

Commercial areas like **MG Road and Koramangala** show consistently high noise due to:

- Heavy traffic
- Frequent honking

- High pedestrian activity

Industrial areas like **Harohalli** also have high noise due to:

- Machinery
- Transportation of goods
- Continuous industrial operations

Relatively quieter areas like **Cubbon Park** still experience noise fluctuations.

This shows that noise pollution affects **all parts of the city**.

Noise levels are highest in the **evening**, when activity peaks.

Morning and afternoon levels are also high, indicating **all-day exposure**.

When compared with standards by the World Health Organization:

- Levels above **70 dB** can harm health
- Many readings exceed safe limits

Noise levels between **80–90 dB** pose serious risks such as:

- Hearing loss
- Stress
- Sleep disturbance

The results also show:

- Weak enforcement of regulations
- Poor urban planning and zoning

Rapid urbanization contributes significantly to rising noise levels.

Mobile-based noise detection tools proved useful for data collection.

They may not be highly precise but are effective for large-scale monitoring.

monitoring and policymaking.

Noise pollution in Bangalore is increasing due to:

- Urban growth
- Transportation expansion
- Industrial activity

Sustainable planning and strict regulations are necessary to reduce noise pollution.

Conclusion & Discussion

This study on noise pollution in Bangalore helps us understand how city activities increase noise levels across different areas. The study collected data from multiple locations and shows that noise pollution is present everywhere:

- Residential areas
- Commercial areas
- Industrial zones
- Even relatively quiet places

The study found that noise levels are not uniform across the city.

Commercial areas such as MG Road and Koramangala have high noise levels due to:

- Heavy traffic

- Frequent honking
- High human activity

Industrial areas like Harohalli also experience high noise levels because of:

- Machinery
- Industrial operations

In quieter areas like Cubbon Park and some residential zones, noise levels fluctuate.

This shows that noise pollution is not limited to specific locations—it affects the entire city.

Noise pollution is a major issue in Bangalore. It is important to examine noise levels in all areas, not only commercial or industrial zones.

The study clearly shows that:

- Noise levels are high across Bangalore
- Noise levels vary depending on location and activity

Noise levels increase significantly during peak activity times, especially in the evening. However, morning and afternoon levels are also high.

This means:

- Noise pollution exists throughout the day
- Cities remain constantly active
- Noise control measures are insufficient

Another important observation is that recorded noise levels often exceed environmental limits.

This raises serious concerns for both human health and the environment.

Prolonged exposure to high noise levels can cause:

- Stress
- Hearing loss
- Sleep disturbances
- Reduced productivity

Therefore, it is necessary to take action to reduce noise pollution and improve urban living conditions.

The study also highlights the importance of data in understanding environmental issues. Using mobile-based tools for noise detection and statistical analysis helped:

- Collect large amounts of real-time data
- Analyze patterns effectively

This approach provides a realistic understanding of noise trends.

It can be used in the future for:

- Monitoring environmental conditions

- Developing better policies

In conclusion, noise pollution in Bangalore is a growing environmental problem caused by:

- Rapid urbanization
- Increased transportation
- Industrial activities

To address this issue, it is necessary to:

- Enforce noise regulations strictly
- Improve urban planning
- Increase public awareness

Reducing noise pollution requires sustainable planning and collective effort.

This study contributes to environmental understanding by:

- Providing practical insights
- Highlighting the need for sustainable solutions

It emphasizes the importance of managing noise pollution effectively in large cities like Bangalore.

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